

YUSUF BAIG

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EDUCATION

New York University (Courant Institute of Mathematical Sciences), New York City, NY Sep 2024 - May 2026
Master of Science in Computer Science
Coursework: Operating system, Machine Learning, NLP, Language Vision Models, Algorithms

Manipal Institute of Technology, Udupi, India July 2018 - May 2022
Bachelor of Technology in Information and Technology, Minor in Computational Intelligence
Coursework: Linear Algebra, Data Structures, Statistics, Machine Learning, Computer Vision

PUBLICATIONS

- Khatri, M., **Yusuf, M.**, Kumar, Y., Shah, R. R., & Kumaraguru, P. (2023). *Exploring Graph Neural Networks for Indian Legal Judgment Prediction*. *arXiv:2310.12800*. *arXiv 2023*
- **Yusuf, M.**, Surana, P., & Sharma, C. (2022). *HindiWSD: A package for word sense disambiguation in Hinglish & Hindi*. *WILDRE-6 ACL 2022*
- **Yusuf, M.**, Surana, P., Gupta, G., & Ramesh, K. (2022). *Curb Your Carbon Emissions: Benchmarking Carbon Emissions in Machine Translation*. *arXiv:2109.12584*.
- Surana, P., **Yusuf, M.**, & Singh, S. (2021). *Severity Classification of Mental Health Related Tweets*. *ICML CA2MH 2021 and IEEE DISCOVER 2021*
- **Yusuf, M.**, Surana, P., Gupta, G., & Ramesh, K. (2021). *Assessing the Carbon Intensity of Models Across Different Languages*. *ICML WiML 2021*

EXPERIENCE

New York University **New York City, New York | April 2025 – Present**
Research Assistant **Advisor: Laure Zanna**

- Designed and trained a GPU-accelerated Swin Transformer surrogate model in PyTorch on 30 years of high-fidelity ocean wave forecasts to emulate sea-state predictions (heights, periods, directions) at a fraction of traditional computational cost.

Multimodal Digital Media Analysis (MIDAS) Lab, IIIT Delhi **Delhi, India | February 2022 – August 2024**
Research Assistant – Natural Language Processing **Advisor: Dr. Jessy Li, Dr. Rajiv Ratn Shah**

- Implemented GNNs for Legal Judgment Prediction, leveraging node classification techniques with advanced XLNet-based embeddings to model judicial case structures, achieving a macro F1 score of 75%.
- Developed and evaluated zero-shot Legal Judgment Prediction methods using multiple Large Language Models leveraging techniques such as Chain of Thought and metadata integration to simulate judicial reasoning processes, achieving a 10% improvement in F1 score compared to traditional approaches.

Carl Zeiss R&D **Bangalore, India | January 2022 – June 2024**
Software Engineer – Image Processing and Backend **Supervisor: Dr. Daniel Weiss**

- Developed advanced Computed Tomography (CT) image processing Python packages from scratch for aerospace clients, specializing in calculating 3D volume Contrast Discrimination Function and Modulation Transfer Function to enhance imaging precision.
- Engineered and optimized complex algorithms for pixel intensity adjustments, streamlining workflows and reducing processing time by 33% (saving 10 hours in a 30-hour project timeframe) during the image processing phase.
- Experienced in building scalable Client-Server Applications utilizing gRPC and Windows Communication Framework (WPF), enabling efficient and high-performance communication between distributed systems.

SKILLS

Languages	Python, C++/C, C#, MATLAB, SQL, Java, HTML, CSS, $\LaTeX$, JavaScript
ML Frameworks	PyTorch, HuggingFace, pandas, LangChain, NumPy, scikit-learn, seaborn, matplotlib
Frameworks	HPC, Git, GitHub, Docker, Flask, Jira, BitBucket, .NET, MongoDB, AWS, Linux

PROJECTS

Solving the ARC challenge

- Developed CogniLLM: a 14 B-parameter vision-language Transformer with custom grid tokenization, LoRA-based cognitive priors, dual-stage transduction-induction solver, and test-time adaptation with ensemble voting, doubling GPT-4o to 29 % ARC task accuracy on a single A100.

Mathematical Pitfalls and Logical Evaluation Score for Mathematical Reasoning

- Designed the MAPLE score, a holistic metric to assess reasoning misalignment in LLMs by integrating error rates, redundancy, and validity.

TEACHING EXPERIENCE

CSCI-UA 381 Programming Tools for the Data Scientist January 2025 - Present

- Assisting Dr. Hasan Aljabbouli with mentoring and grading for the undergraduate course Programming Tools for the Data Scientist Spring 2025 semester at New York University.